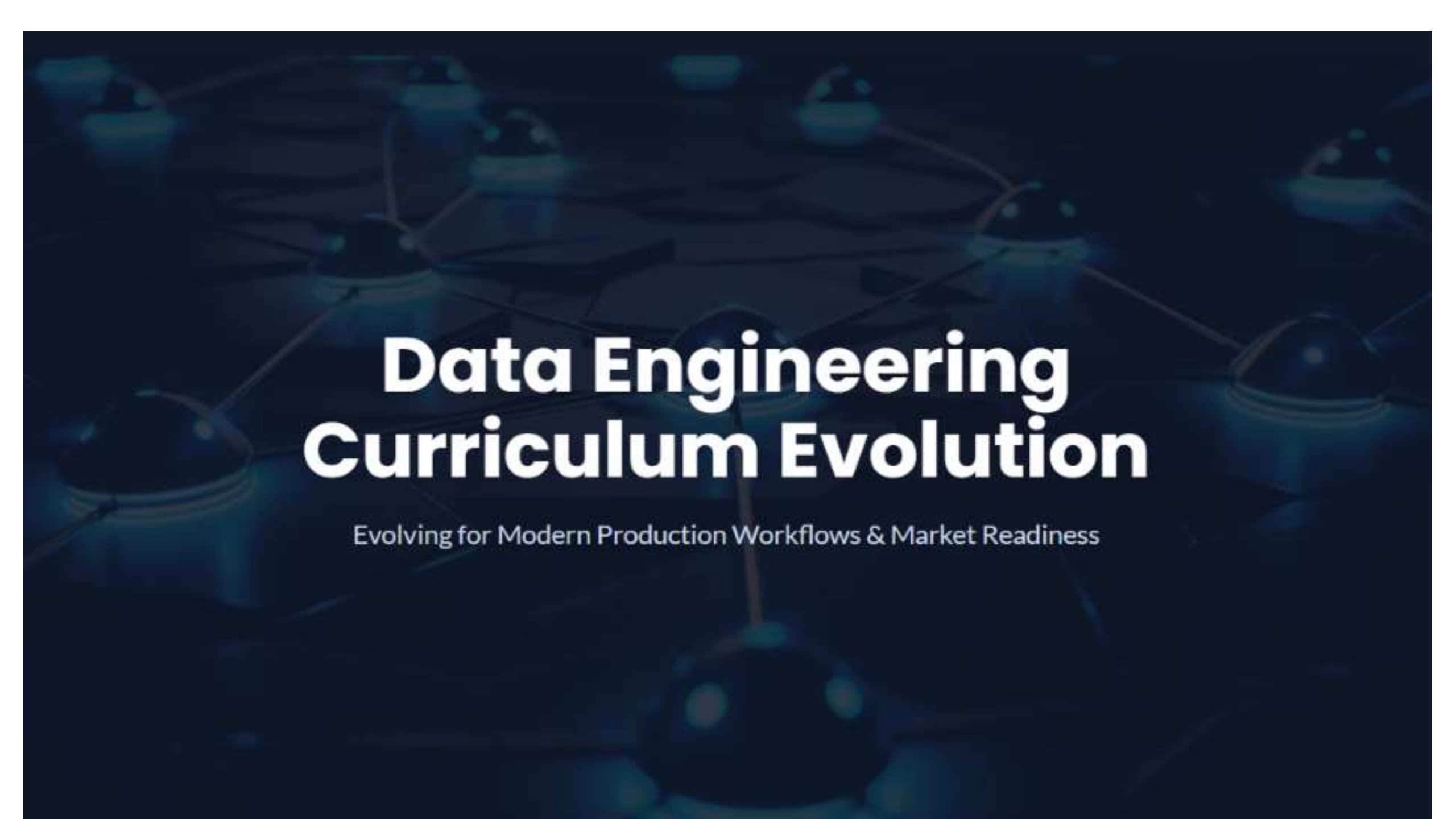


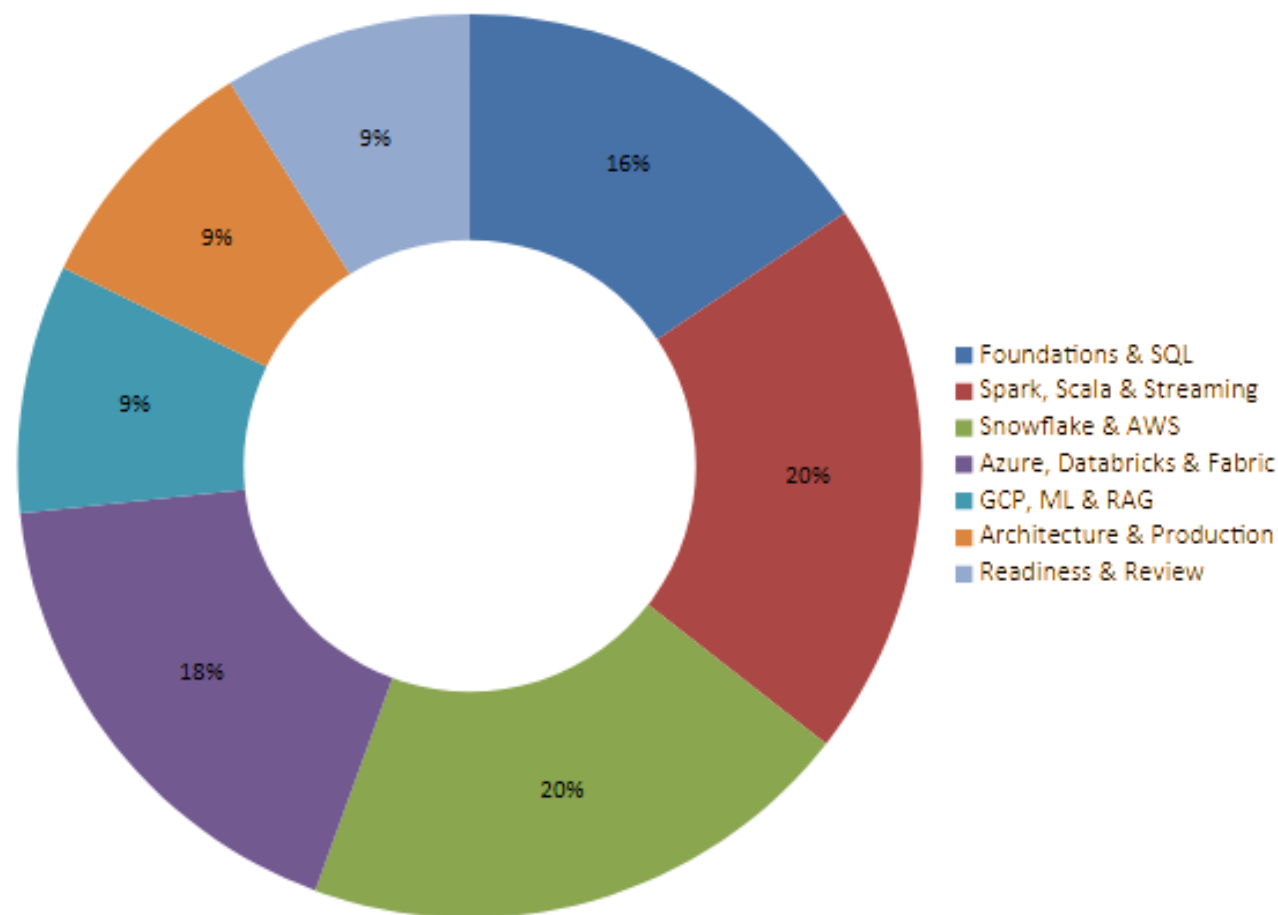


# **Data Engineering Curriculum Evolution**

Evolving for Modern Production Workflows & Market Readiness

# Training Bandwidth Allocation

45-day curriculum — approximate allocation by major learning area



## What the allocation shows

Core engineering + distributed processing remain the largest investment.

Cloud coverage is broad, but tied to practical Data Engineering workflows.

ML/RAG are included as AI-ready Data Engineering skills—not as a separate ML program.

Architecture, production break/fix and interview readiness are explicitly protected.

# Week 1: Engineering Foundations

| Day   | Topic                         | Coverage                                                                                                  |
|-------|-------------------------------|-----------------------------------------------------------------------------------------------------------|
| Day 1 | Data Engineering Fundamentals | Data lifecycle, ETL/ELT, batch vs streaming, OLTP/OLAP, data lake/warehouse, idempotency, error handling. |
| Day 2 | Linux & Shell Scripting       | Filesystem, permissions, grep/awk/sed, processes, cron, shell automation and operational logs.            |
| Day 3 | Python + Pandas               | Functions, exceptions, logging, file I/O, JSON/CSV, DataFrames, filtering, joins, and small ETL.          |
| Day 4 | Git Workflow                  | Branching, commits, PRs, merge conflicts, code review, repository discipline and release basics.          |
| Day 5 | CI/CD for Data Eng.           | Build/test/deploy flow, pipeline YAML, PR validation, environments, secrets, rollback and patterns.       |

## Week 2: SQL & Distributed Systems

| Day    | Topic                      | Coverage                                                                                                 |
|--------|----------------------------|----------------------------------------------------------------------------------------------------------|
| Day 6  | SQL Fundamentals           | Joins, aggregations, CTEs, window functions, subqueries, query plans, indexing and interview prep.       |
| Day 7  | Data Warehouse Concepts    | Facts, dimensions, grain, surrogate keys, SCD1/SCD2, star/snowflake schemas, and medallion architecture. |
| Day 8  | Distributed Systems + HDFS | Horizontal scaling, fault tolerance, CAP basics, HDFS blocks/replication, NameNode/DataNode concepts.    |
| Day 9  | Hive + Spark Intro         | Metastore, tables, partitioning, Parquet/ORC, Hadoop-to-Spark evolution, Spark driver/executors/DAG.     |
| Day 10 | Spark Advanced             | Execution plans, transformations, joins, shuffle, caching, skew, optimization and performance tuning.    |

## Week 3: Scala & Streaming

| Day    | Topic                           | Coverage                                                                                              |
|--------|---------------------------------|-------------------------------------------------------------------------------------------------------|
| Day 11 | Scala Fundamentals              | val/var, functions, collections, functional concepts and reading Spark-oriented Scala code.           |
| Day 12 | Scala Advanced                  | Deep dive into advanced Scala constructs and their application in data engineering workloads.         |
| Day 13 | Incremental Loads & Spark Scala | Collections with Spark patterns, incremental patterns, watermarks, CDC, MERGE and upsert operations.  |
| Day 14 | Kafka + Lab                     | Brokers, topics, partitions, offsets, producer/consumer implementation, retries, and troubleshooting. |
| Day 15 | MongoDB + Spark Streaming       | Document model, indexing basics, Spark streaming integration and semi-structured data patterns.       |

## Week 4: Streaming, Snowflake & dbt

| Day    | Topic                      | Coverage                                                                                              |
|--------|----------------------------|-------------------------------------------------------------------------------------------------------|
| Day 16 | Spark Structured Streaming | Micro-batch, windowing, stateful processing, checkpoints, late data handling and Kafka integration.   |
| Day 17 | Snowflake Introduction     | Architecture, virtual warehouses, storage/compute separation, caching, Time Travel and cloning.       |
| Day 18 | Snowflake Advanced         | RBAC, masking, row access, performance tuning, pruning, clustering, sizing and cost controls.         |
| Day 19 | dbt (Data Build Tool)      | Sources, models, DAG, materializations, tests, incremental models, snapshots, docs and lineage.       |
| Day 20 | AWS Intro + IAM + S3       | Regions, IAM users/roles/policies, least privilege, S3 versioning, lifecycle, and data-lake patterns. |

## Week 5: AWS Data Engineering

| Day    | Topic                    | Coverage                                                                                             |
|--------|--------------------------|------------------------------------------------------------------------------------------------------|
| Day 21 | EC2 + Lambda             | Compute basics, security groups, serverless event-driven processing, triggers, retries, and logging. |
| Day 22 | AWS Glue                 | Glue crawlers, catalog, jobs, Spark ETL integration, schema handling, and job bookmarks.             |
| Day 23 | EMR + Redshift Intro     | EMR architecture, operational patterns; Redshift architecture, COPY, distribution/sort strategies.   |
| Day 24 | Redshift + Kinesis + RDS | Warehouse optimization, streaming ingestion with Kinesis, and relational-source patterns with RDS.   |
| Day 25 | Step Functions + DMS     | Workflow orchestration, state machines, retries; database migration, replication, and CDC patterns.  |

## Week 6: Platform Engineering & Databricks

| Day    | Topic                      | Coverage                                                                                           |
|--------|----------------------------|----------------------------------------------------------------------------------------------------|
| Day 26 | Terraform (IaC)            | Infrastructure as Code, providers, resources, state, reusable modules, and deployment controls.    |
| Day 27 | Apache Airflow             | DAGs, operators, dependencies, sensors, XCom, retries, catchup, backfill and failure handling.     |
| Day 28 | Azure Entra ID + ADLS Gen2 | Azure identity/RBAC, storage accounts, hierarchical namespaces, security, and data-lake patterns.  |
| Day 29 | Databricks Intro           | Workspace/compute, Spark integration, jobs, notebooks, Delta fundamentals, medallion architecture. |
| Day 30 | Databricks Advanced        | Delta MERGE, schema enforcement/evolution, optimization, performance, cost, and security patterns. |

## Week 7: Azure Analytics & Microsoft Fabric

| Day    | Topic                    | Coverage                                                                                               |
|--------|--------------------------|--------------------------------------------------------------------------------------------------------|
| Day 31 | ADF + Unity Catalog      | ADF pipelines, activities, triggers plus Databricks governance concepts, catalogs, schemas, lineage.   |
| Day 32 | Microsoft Fabric (Day 1) | OneLake, Lakehouse, Data Warehouse, Data Factory and overall Fabric architecture breakdown.            |
| Day 33 | Microsoft Fabric (Day 2) | Direct Lake, mirroring, governance/integration patterns and deep hands-on application.                 |
| Day 34 | Synapse + Event Hubs     | Synapse serverless/dedicated/Spark concepts plus Azure streaming/event ingestion with Event Hubs.      |
| Day 35 | Presentation             | Trainee presentations and review of architectural concepts covered throughout the core platform weeks. |

## Week 8: GCP, ML & AI-Ready Data Eng.

| Day    | Topic                    | Coverage                                                                                                 |
|--------|--------------------------|----------------------------------------------------------------------------------------------------------|
| Day 36 | GCP: BigQuery + Pub/Sub  | Cloud Storage, BigQuery architecture, partitioning, clustering, Pub/Sub topics and streaming patterns.   |
| Day 37 | Dataflow + Dataproc      | Apache Beam pipelines, transforms; Dataproc Spark/Hadoop and service-selection comparison.               |
| Day 38 | ML Fundamentals & MLFlow | ML lifecycle, features/labels, train/test, MLFlow experiments, runs, metrics, and model registry.        |
| Day 39 | RAG & Vector Data Eng.   | Chunking, embeddings, vector stores, ingestion pipelines, metadata, lineage, and retrieval architecture. |
| Day 40 | Production Break/Fix     | Deliberately break pipelines, detect incidents, diagnose root causes, recover, and document RCAs.        |

## Week 9: Modern Architecture & Readiness

| Day    | Topic                          | Coverage                                                                                                |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Day 41 | Open Table Formats & Contracts | Iceberg/Hudi concepts, ACID, schema evolution, data catalogs, PII, and producer-consumer contracts.     |
| Day 42 | Data Mesh & MDM                | Domain ownership, data products, hubs-links-satellites, golden records, and architecture awareness.     |
| Day 43 | Kubernetes for Data Eng.       | Containers, pods, deployments, services, and where Kubernetes appears in managed data platforms.        |
| Day 44 | Flexible Extension             | Time reserved for advanced mock interviews, specific topic review, and technical interview preparation. |
| Day 45 | Marketing Readiness            | Final preparations for the job market, resume review, behavioral alignment, and graduation activities.  |



## **Questions & Discussion**